

Unit 6 Review: Similarity Transformations

NO WORK; NO CREDIT

1. Which statement would prove there is a similarity transformation between two triangles?

- A. If two angles of one triangle are proportional to two angles of another triangle, then the triangles are similar.
- B. If two sides of one triangle are congruent to two sides of another triangle, then the triangles are similar.
- C. If two sides of one triangle are proportional to two sides of another triangle, then the triangles are similar.
- D. If two angles of one triangle are congruent to two angles of another triangle, the two triangles are similar.

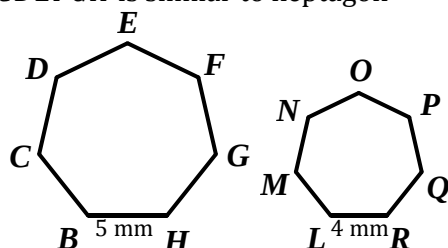
2. There are two rectangles.

- Rectangle $WXYZ$ has a length of 21 inches and a width of 7 inches.
- For rectangle $RSTU$, the length is three times its width.

Which statement will **always** be true of the relationship between rectangle $WXYZ$ and rectangle $RSTU$?

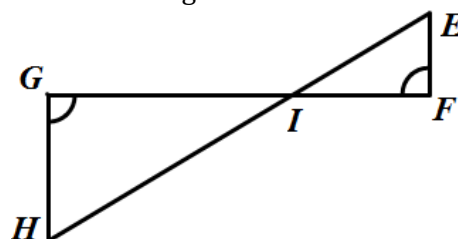
- A. Rectangle $WXYZ$ and rectangle $RSTU$ are congruent.
- B. Rectangle $WXYZ$ and rectangle $RSTU$ are similar.
- C. The area of rectangle $RSTU$ is three times the area of rectangle $WXYZ$.
- D. The areas of rectangle $WXYZ$ and rectangle $RSTU$ are the same.

3. Heptagon $BCDEFGH$ is similar to heptagon $LMNOPQR$.



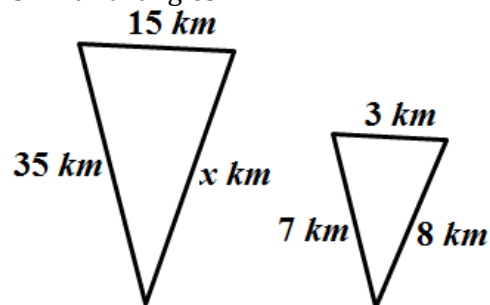
What must be true of the measure of angle B ?

4. The two triangles shown are similar.



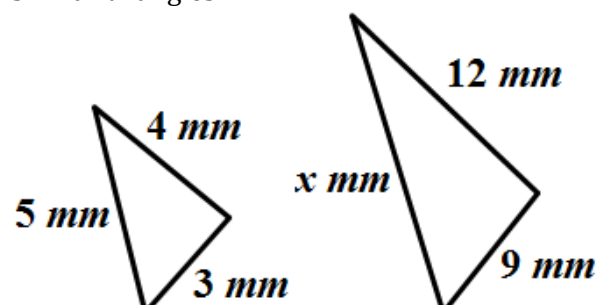
Which angle has the same measure as $\angle FEI$?

5. The diagram shows the side lengths of two similar triangles.



What is the value of x , in kilometers?

6. The diagram shows the side lengths of two similar triangles.

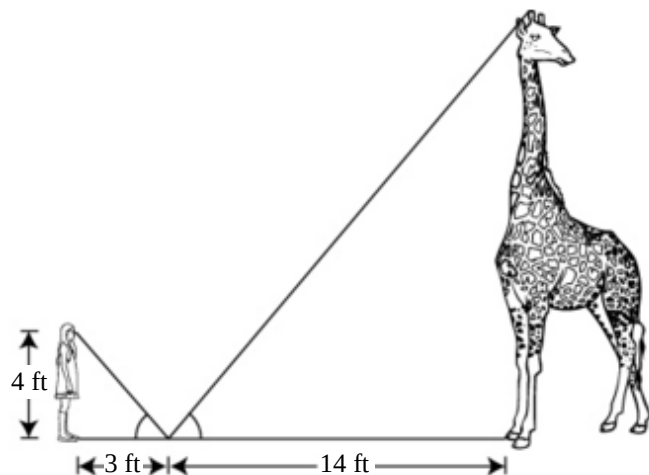


What is the value of x , in millimeters?

Unit 6 Review: Similarity Transformations

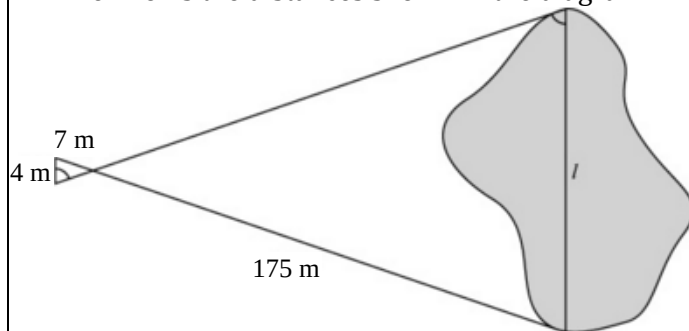
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7. Ramona wants to find the height of the giraffe statue outside the library. She places a mirror on the ground and walks backward until she can see the giraffe's head in the mirror. She knows the measurements shown in the picture.



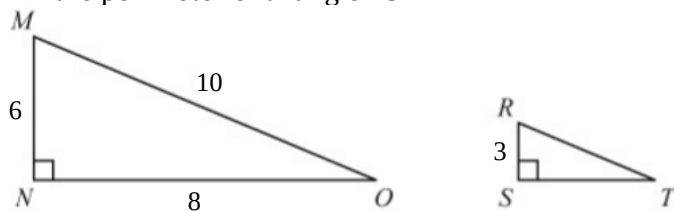
About how tall is the statue? (Round to the nearest whole value.)

8. Mr. Smith wants to find the length, l , of a pond. He knows the distances shown in the diagram.

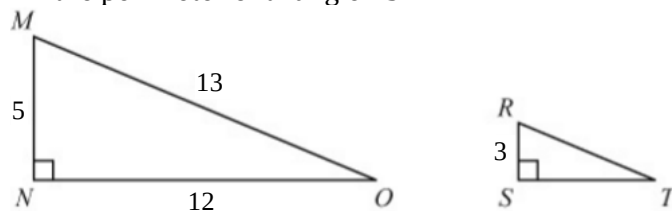


What is the length, in meters, of the pond?

9. If triangle MNO is similar to triangle RST , what is the perimeter of triangle RST ?



10. If triangle MNO is similar to triangle RST , what is the perimeter of triangle RST ?



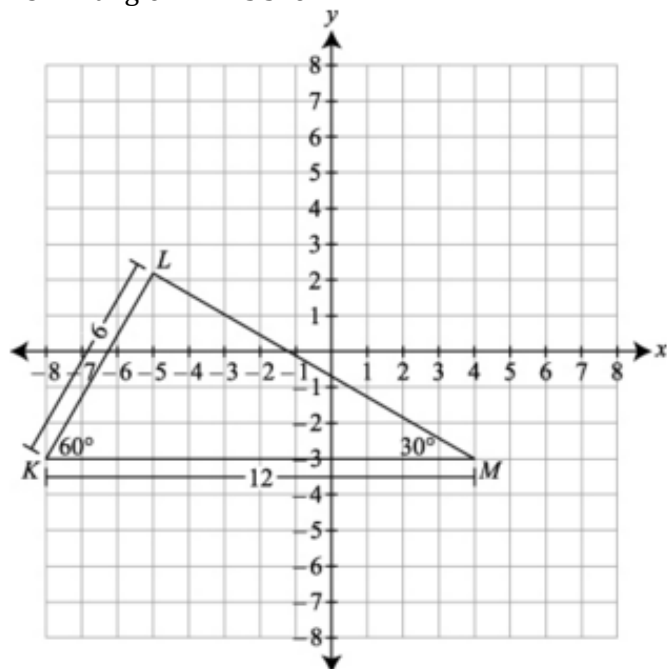
Unit 6 Review: Similarity Transformations

NO WORK; NO CREDIT

11. At a certain time of day, a tree has a shadow 60 feet long. At the same time, a 5-foot-tall man stands next to the tree, and his shadow is 15 feet long. What is the height of the tree, in feet?

12. At a certain time of day, a tree has a shadow 50 feet long. At the same time, a 6-foot-tall man stands next to the tree, and his shadow is 20 feet long. What is the height of the tree, in feet?

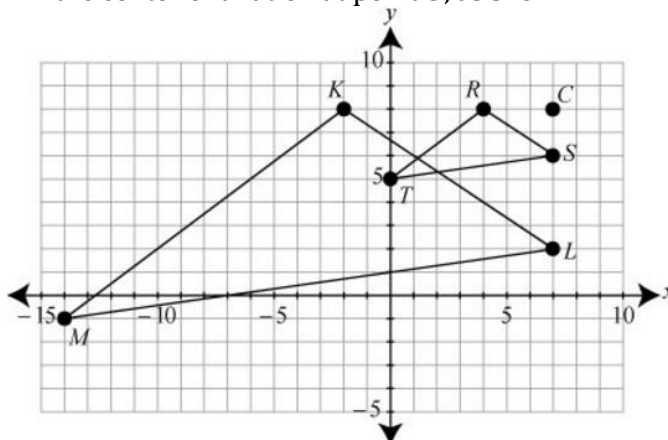
13. Triangle KLM is shown.



Triangle KLM is dilated by a factor of $\frac{1}{3}$ and translated 4 units to the right. Which statement about the image $\Delta K'L'M'$ is true?

- A. $K'L' = 18$ units
- B. $K'M' = 12$ units
- C. $m\angle K' = 30^\circ$
- D. $m\angle M' = 30^\circ$

14. Triangle RST is a dilation of triangle KLM with the center of dilation at point C , as shown.

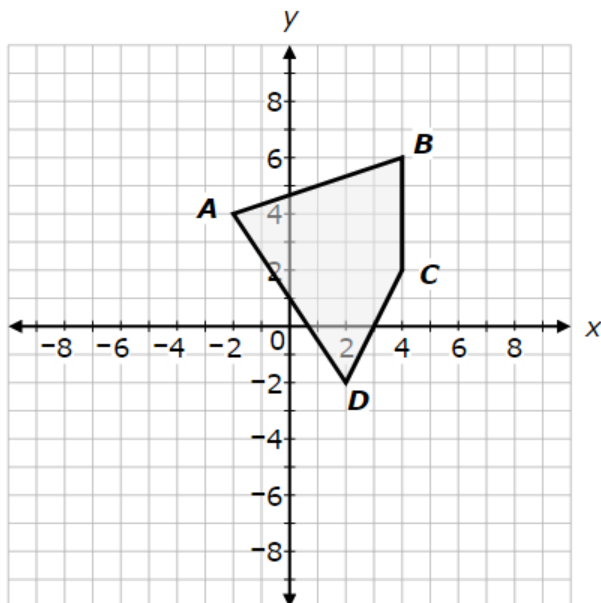


What is the scale factor of this dilation?

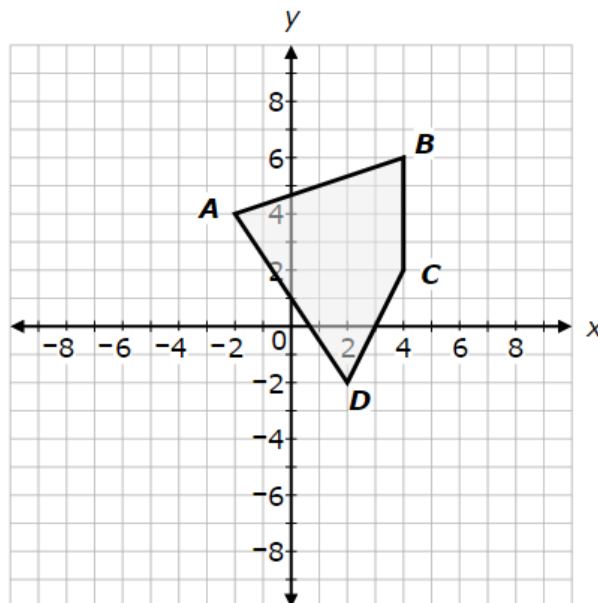
Unit 6 Review: Similarity Transformations

NO WORK; NO CREDIT

15. Quadrilateral $ABCD$ is shown on the coordinate grid. The quadrilateral is dilated by a factor of 0.5 about the origin to create $A'B'C'D'$. Use the grid to create $A'B'C'D'$.



16. Quadrilateral $ABCD$ is shown on the coordinate grid. The quadrilateral is dilated by a factor of 1.5 about the origin to create $A'B'C'D'$. Use the grid to create $A'B'C'D'$.



17. A rectangle is dilated on the coordinate plane. The area of the dilated rectangle is 4 times the area of the original rectangle. What is the scale factor of the dilation?

18. A rectangle is dilated on the coordinate plane. The area of the dilated rectangle is 25 times the area of the original rectangle. What is the scale factor of the dilation?